Vehicle Lease Contract Parser - Python Code

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# Install dependency
!pip install PyPDF2

# Upload PDF
from google.colab import files
uploaded = files.upload()

# SLA fields structure
sla_fields = {
    "apr_percent": None,
    "term_months": None,
    "monthly_payment": None,
    "down_payment": None,
    "residual_value": None,
    "mileage_allowance": None,
    "mileage_overage_fee": None,
    "early_termination_fee": None,
    "purchase_option_price": None,
    "insurance_requirements": None,
    "maintenance_responsibilities": None,
    "warranty_summary": None,
    "late_fee_policy": None
}

# Extract text from PDF
import PyPDF2

file_name = list(uploaded.keys())[0]
with open(file_name, 'rb') as f:
    reader = PyPDF2.PdfReader(f)
    text = ""
    for page in reader.pages:
        text += page.extract_text()

print(text[:1000])

# APR Extraction
import re

apr_patterns = [
    r"Annual Percentage Rate\s*[:\-]?\s*([\d.]+)\s*%",
    r"APR\s*[:\-]?\s*([\d.]+)\s*%",
    r"Interest Rate\s*[:\-]?\s*([\d.]+)\s*%"
]

for pattern in apr_patterns:
    match = re.search(pattern, text, re.IGNORECASE)
    if match:
        sla_fields["apr_percent"] = match.group(1)
        break

# Monthly Payment
monthly_match = re.search(
    r"(Monthly Payment|Payment Amount)[^\d]\$?([\d,\.]+)",
    text, re.IGNORECASE
)
if monthly_match:
    sla_fields["monthly_payment"] = monthly_match.group(2)

# Term
term_match = re.search(r"(\d+)\s(months|month term)", text, re.IGNORECASE)
if term_match:
    sla_fields["term_months"] = term_match.group(1)

# Down Payment
down_match = re.search(
    r"(Down Payment|Initial Payment)[^\d]\$?([\d,\.]+)",
    text, re.IGNORECASE
)
if down_match:
    sla_fields["down_payment"] = down_match.group(2)

# Residual Value
residual_match = re.search(
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    r"(Residual Value|End Value)[^\d]\$?([\d,\.]+)",
    text, re.IGNORECASE
)
if residual_match:
    sla_fields["residual_value"] = residual_match.group(2)

# Mileage Allowance
mileage_match = re.search(
    r"(Mileage Allowance|Annual Mileage)[^\d](\d+)",
    text, re.IGNORECASE
)
if mileage_match:
    sla_fields["mileage_allowance"] = mileage_match.group(2)

# Mileage Overage Fee
overage_match = re.search(
    r"(Overage Fee|Excess Mileage)[^\d]\$?([\d,\.]+)",
    text, re.IGNORECASE
)
if overage_match:
    sla_fields["mileage_overage_fee"] = overage_match.group(2)

# Save SLA fields
import json

with open("contract_sla.json", "w") as f:
    json.dump(sla_fields, f, indent=4)

# Decode VIN
import requests

vin = "1HGCM82633A004352"
url = f"https://vpic.nhtsa.dot.gov/api/vehicles/DecodeVin/{vin}?format=json"

response = requests.get(url)
data = response.json()

vehicle_info = {"make": None, "model": None, "year": None}

for item in data["Results"]:
    if item["Variable"] == "Make":
        vehicle_info["make"] = item["Value"]
    elif item["Variable"] == "Model":
        vehicle_info["model"] = item["Value"]
    elif item["Variable"] == "Model Year":
        vehicle_info["year"] = item["Value"]

# Save combined data
contract_data = {
    "sla_fields": sla_fields,
    "vehicle_info": vehicle_info
}

with open("contract_with_vehicle.json", "w") as f:
    json.dump(contract_data, f, indent=4)

# Field explanations
explanations = {
    "apr_percent": "APR is the yearly interest charged.",
    "term_months": "Length of the lease in months.",
    "monthly_payment": "Amount paid every month.",
    "down_payment": "Upfront payment.",
    "residual_value": "Estimated value at lease end."
}

for field, value in sla_fields.items():
    print(field, ":", value)
    print(explanations.get(field, "No explanation available"))
```